Λύσεις Ενδιάμεσης Εξέτασης

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Άσκηση 1) α) Στο σημεία Α η κλίση είναι αρνητική: $\frac{dU}{dx} > 0 \Rightarrow F = -\frac{dU}{dx}$
Άρα η κατεύθυνση της δύναμης είναι προς τα δεξιά. ($F > 0$).

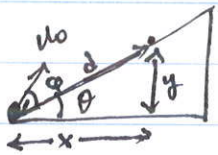
β) $\frac{dU}{dx} > 0 \Rightarrow F < 0$. , γ) Εκεί όπου $U(x)$ είναι ελάχιστη $x = 0,75m$

δ) $\frac{dU}{dx} = 0 \Rightarrow F = 0$. ε) Πρέπει να βρίσκεται στην ίδιο οριζόντια ευθεία με το Α : $x \approx 2,25m$

ζ) Ευσταθή ισορροπία: $x = 0,75m$, $x = 1,75m$

η) Ασταθή ισορροπία: $x = 1,25m$ (σημείο c).

Άσκηση 2).



i) $d^2 = x^2 + y^2$, $x = d\cos\vartheta$, $y = d\sin\vartheta$

$a_x = 0$, $u_x = u_0 \cos(\theta+\varphi)$, $x = u_0 \cos(\theta+\varphi)\, t$

$a_y = -g$, $u_y = u_0 \sin(\theta+\varphi) - gt$, $y = u_0 \sin(\theta+\varphi).t - \frac{1}{2} g t^2$

Αν $x = d\cos\vartheta \Rightarrow t = t_{on} \Rightarrow d\cos\vartheta = u_0 . \cos(\theta+\varphi).\, t \Rightarrow t = \frac{d\cos\vartheta}{u_0 \cos(\theta+\varphi)}$.

Για $t = t_{on}$ $y = d\sin\vartheta$

$$\Rightarrow d\sin\vartheta = u_0 \sin(\theta+\varphi).\frac{d\cos\vartheta}{u_0\cos(\theta+\varphi)} - \frac{1}{2} g \left(\frac{d\cos\vartheta}{u_0\cos(\theta+\varphi)}\right)^2$$

$$\Rightarrow \sin\vartheta = \frac{\sin(\theta+\varphi)\cos\vartheta}{\cos(\theta+\varphi)} - \frac{1}{2} g \frac{d\cos^2\vartheta}{u_0^2 \cos^2(\theta+\varphi)}$$

$$\Rightarrow d = \frac{2u_0^2 \cos^2(\theta+\varphi)}{g\cos^2\vartheta}\left(\frac{\sin(\theta+\varphi)\cos\vartheta}{\cos(\theta+\varphi)} - \sin\vartheta\frac{\cos(\theta+\varphi)}{\cos(\theta+\varphi)}\right).$$

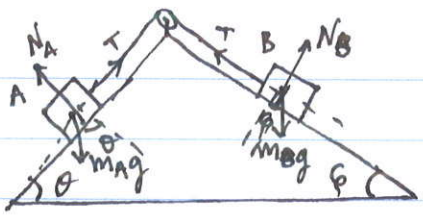
$$= \frac{2u_0^2}{g}\frac{\cos(\theta+\varphi)}{\cos^2\vartheta}\left(\sin(\theta+\varphi)\cos\vartheta - \sin\vartheta\cos(\theta+\varphi)\right)$$

$$d = \frac{2u_0^2}{g}\frac{\cos(\theta+\varphi)\sin\varphi}{\cos^2\vartheta}.$$

ii) $\frac{\partial d}{\partial \varphi} = 0 \Rightarrow \frac{\partial d}{\partial \varphi} = \frac{2u_0^2}{g\cos^2\vartheta}\left(-\sin(\theta+\varphi)\sin\varphi + \cos(\theta+\varphi)\cos\varphi\right) = \frac{2u_0^2}{g\cos^2\vartheta}\cos(\theta+2\varphi)$

$\Rightarrow \cos(\theta+2\varphi) = 0 \Rightarrow \theta + 2\varphi = \pi/2 \Rightarrow \boxed{\varphi = \frac{\pi}{4} - \frac{\theta}{2}}$

Άσκηση 3)



$\mu_K = 0,4$

$m_A = 10\,kg$, $m_B = 50\,kg$

$\theta = 30°$, $\varphi = 45°$

i) $m_B > m_A$ $\quad m_A g\sin\theta = 10 \cdot 9,81 \cdot \frac{1}{2} = 49,0N$

$m_B g\sin\varphi = 50 \cdot 9,81 \cdot \frac{\sqrt{2}}{2} = 347N$

$F_{KA} = \mu_K N_A = \mu_K \cdot m_A g\cos\theta = 0,4 \cdot 10 \cdot 9,81 \frac{\sqrt{3}}{2} = 34,0N$

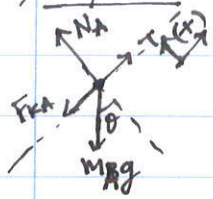
$F_{KB} = \mu_K N_B = \mu_K m_B g\cos\varphi = 0,4 \cdot 50 \cdot 9,81 \frac{\sqrt{2}}{2} = 139,0N$

Έστω ότι θα κινηθεί δεξιά $\Rightarrow -F_{KA} + m_A g\sin\theta - F_{KB} + m_B g\sin\varphi > 0$

Έστω ότι θα κινηθεί αριστερά $\Rightarrow m_A g\sin\theta - F_{KA} - F_{KB} - m_B g\sin\varphi < 0$

} Θα κινηθεί δεξιά.

ii) Σώμα Α



Σώμα Β



$N_A - m_A g\cos\theta = 0$

$T - F_{KA} - m_A g\sin\theta = m_A a$ $\Rightarrow T - \mu_K m_A g\cos\theta - m_A g\sin\theta = m_A a$

$N_B - m_B g\cos\varphi = 0$

$-T - F_{KB} + m_B g\sin\varphi = m_B a$ $\Rightarrow -T - \mu_K m_B g\cos\varphi + m_B g\sin\varphi = m_B a$

Τις προσθέτω

$\Rightarrow m_B g(\sin\varphi - \mu_K\cos\varphi) - m_A g(\sin\theta + \mu_K\cos\theta) = (m_A + m_B)a \Rightarrow a = \frac{m_B(\sin\varphi - \mu_K\cos\varphi) - m_A(\sin\theta + \mu_K\cos\theta)}{m_A + m_B} g$

$a = 2,08\,m/s^2$

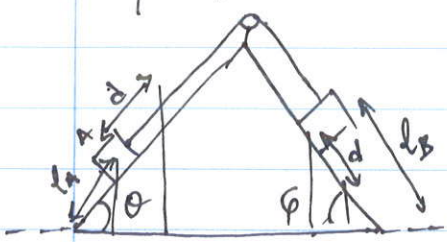
iii) $T = m_A(a + g\sin\theta + \mu_K g\cos\theta) = 104N$

$T = m_B(-a - \mu_K g\cos\varphi + g\sin\varphi) = 104N$ (ok). ✓

iv) α τρόπος). $u = \sqrt{2ad} = \sqrt{2gd\,\frac{m_B(\sin\varphi - \mu_K\cos\varphi) - m_A(\sin\theta + \mu_K\cos\theta)}{m_A + m_B}}$

$= 2,03\,m/s$

β τρόπος). $K_1 + U_1 + W_{other} = K_2 + U_2$



$0 + 0 + m_A g l_A\sin\theta + m_B g l_B\sin\varphi - F_{KA}\cdot d - F_{KB}\cdot d = \frac{1}{2}m_A u^2 + \frac{1}{2}m_B u^2 + m_A g(l_A + d)\sin\theta + m_B g(l_B - d)\sin\varphi$

$\Rightarrow \frac{1}{2}(m_A + m_B)u^2 = -\mu_K m_A g\cos\theta d - \mu_K m_B g\cos\varphi d - m_A g d\sin\theta + m_B g d\sin\varphi$

$\Rightarrow u = \sqrt{2gd\,\frac{m_B(\sin\varphi - \mu_K\cos\varphi) - m_A(\sin\theta + \mu_K\cos\theta)}{m_A + m_B}} = 2,03\,m/s$ (ok) ✓